Churn Radar

Week 1: Data Model, Cleaning & Preprocessing

- **Tasks:** Perform EDA to assess data quality, clean missing values and convert `TotalCharges` to numeric float, encode categorical variables (`Contract`, `InternetService`, `PaymentMethod`), create derived features.
- **Tools:** SQL, Python (Pandas, NumPy, Matplotlib, Seaborn).
- **Deliverables:** Cleaned dataset (`Cleaned_Dataset.csv`) + Data Preprocessing Jupyter Notebook.

Week 2: Analysis Questions & Customer Profiling

- **Tasks:** Answer core business questions regarding churn drivers across demographics (`SeniorCitizen`, `Partner`), contract types, and internet services, perform customer segmentation, calculate financial loss (`Revenue at Risk`).
- **Tools:** SQL, Python (Pandas, Matplotlib, Seaborn).
- **Deliverables:** EDA Notebook with explanatory visualizations + Comprehensive Root-Cause Report.

Week 3: Predictive Analytics & Retention Strategy

- **Tasks:** Build and evaluate ML classification models (Logistic Regression, Random Forest, XGBoost) using metrics like ROC-AUC, Recall, and Precision, perform Feature Importance analysis, formulate targeted retention offers for high-risk customers.
- **Tools:** Python (Scikit-Learn, Pandas).
- **Deliverables:** Trained ML Model Pipeline & Evaluation Metrics + Actionable Retention Strategy Document.

Week 4: Dashboards, Web App & Final Presentation

- **Tasks:** Build executive dashboards in **Power BI**, **Tableau**, and **Looker Studio** showing key KPIs (`Churn Rate`, `Revenue at Risk`); develop a real-time **Streamlit** Web App for agents to input customer data and receive retention offers; prepare final presentation slides and executive report.
- **Tools:** Power BI, Tableau, Looker Studio, Python (Streamlit), SQL.
- **Deliverables:** Interactive Multi-Platform Dashboards + Streamlit Web App + Final Project Report & Slide Deck.